A Dyadic Unique-Maximum Conjecture for Adjacent-Difference Fibers

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Preprint, version 2 | 14 November 2026

Abstract

We conjecture that a recursive ordered difference word is the unique largest permutation fiber at every order 2^k. We give a complete hand-checkable four-label model, an exact endpoint-matrix recurrence, and rigidity under a prescribed dyadic distance multiset. At order 32 an archived exhaustive 16-label energy certificate proves uniqueness among all records with one odd entry. Within this class, a path-spectrum bridge bound improves the upper bound for competitors with a half different from D_{16} to 47996, below the candidate's 48410. Parity-run balance and a two-odd-entry contraction identify the additional data needed beyond one crossing. The unrestricted order-32 inequality and the all-orders conjecture remain open.

Preprint; not externally peer reviewed. AI use and verification are described at the end.

Introduction and research strategy

The dyadic assertion was extracted from the general maximum-ambiguity problem rather than imposed on all orders. Complete small searches produced unique optimal records at orders 2, 4 and 8 following one recursion; the later order-16 search retained both that shape and uniqueness. The notes distinguish these stages from the subsequent calculation of much larger candidate fibers. Large candidate counts were not evidence of global optimality by themselves.

The recursion has an interpretable mechanism. In D_{2n}, all steps are even except the central step of length one. Any realization must exhaust one parity class, cross once, and exhaust the other. This explains why the candidate can be counted by endpoint matrices. It does not establish that competing records must organize themselves in the same way. The distinction is crucial at order 32, where competitors with several odd jumps remain uncontrolled.

Research strategy. We give a hand-checkable four-label model, the exact endpoint recurrence, and a rigidity theorem when the entire distance histogram is prescribed. For the one-odd-jump class we combine a finite endpoint-energy certificate with a uniform bridge estimate. An exact path-spectrum calculation sharpens the bound on all competitors with a non-dyadic half to 47996, below the candidate's 48410. Finally a two-odd-jump decomposition identifies the additional disjoint-path data a global argument must control. These are successive structural reductions, not a proof that the remaining class is negligible.

1 The conjecture and its research context

For $\pi \in S_n$, let $\Delta(\pi) = (|\pi_{i+1} - \pi_i|)_{i=1}^{n-1}$ and $A_n(d) = \#\{\pi : \Delta(\pi) = d\}$. The word is ordered, and permutations are not identified under reflection or reversal. Define

$$D_1 = (), \qquad D_{2n} = (2D_n, 1, 2D_n) \quad (n \text{ a power of two}).$$

Equivalently, $(D_n)_i = n/2^{v_2(i)+1}$ for $1 \le i < n$.

Conjecture 1.1. For every $n = 2^k$, $k \ge 1$, and every word $d \in \{1, \dots, n-1\}^{n-1}$,

$$A_n(d) \le A_n(D_n), \qquad A_n(d) = A_n(D_n) \iff d = D_n.$$

There is no hypothesis that a competitor has a dyadic multiset, a parity decomposition, or a small number of odd entries.

Reconstruction from ordered absolute differences is already studied by De Biasi [1]. Bulteau–Giraudo–Vialette study the same mapping in a different extremal statistic, additive disorder [2]. Neither introducing the mapping nor optimizing the sum of its entries is the proposed contribution. A focused literature search through 14 November 2026 did not locate this exact unique-maximum assertion in the checked sources; no exhaustive priority claim is made.

1.1 A hand-checkable four-label model

At order four, uniqueness can be checked without a search program. Complementation pairs every permutation with one whose first label lies in $\{1, 2\}$. There are exactly twelve such representatives. Grouping them gives the complete table:

Record	Fiber	Representatives with first label at most 2
$(1,1,1)$	2	1234
$(1,1,3)$	2	2341
$(1,2,1)$	4	1243, 2134
$(1,2,3)$	2	2314
$(1,3,1)$	2	2143
$(2,1,2)$	6	1324, 1342, 2431
$(2,3,2)$	2	2413
$(3,1,1)$	2	1432
$(3,2,1)$	2	1423

The representative column contains all six orders beginning with 1 and all six beginning with 2, so no other record is omitted. Each representative contributes its distinct complement, proving $M(4) = 6$ and uniqueness of $D_4 = (2,1,2)$. The second-best word $(1,2,1)$ already has a different interval-block mechanism. This small model illustrates why merely exhibiting six realizations is weaker than proving that the full table has no competing row.

2 An exact matrix recurrence for the candidate

Let $E_n(i,j)$ count realizations of D_n starting at i and ending at j, and let C_n have entries

$$C_n(a,b) = \mathbf{1}_{\{b=a \text{ or } b=a-1\}}, \qquad 1 \le a, b \le n.$$

The word D_n is palindromic, so E_n is symmetric. Write $r_i = \sum_j E_n(i,j)$ and $F_n = \sum_i r_i = A_n(D_n)$.

Theorem 2.1. Starting from $E_1 = [1]$, and ordering the $2n$ labels by odd labels followed by even labels,

$$E_{2n} = \begin{pmatrix} 0 & E_n C_n E_n \\ E_n C_n^{\mathsf{T}} E_n & 0 \end{pmatrix}.$$

Consequently

$$F_{2n} = 2\left(\sum_{i=1}^{n} r_i^2 + \sum_{i=1}^{n-1} r_i r_{i+1}\right).$$

Proof. The even differences in the first and second blocks force each block to remain in one parity class. Each block has n vertices, hence exhausts that class. After rescaling, each realizes D_n. A bridge between odd label $2a-1$ and even label $2b$ has length one exactly when $b = a$ or $b = a-1$. Multiplication counts all compatible choices, and the reverse parity order gives the transpose block. Summing the entries proves the scalar formula. □

n	2	4	8	16
F_n	2	6	30	430
n	32	64	128	
F_n	48410	327334518	7794829488660802	

These are exact counts for one candidate, not the values of the global maximum at the last three orders. The supplied program computes the recurrence through 128 and independently enumerates signed walks for D_n through 32.

Proposition 2.2. The limit $\lambda_D = \lim_{k\to\infty} F_{2^k}^{1/2^k}$ exists, and for dyadic $n \geq 2$,

$$\left(\frac{F_n}{n}\right)^{1/n} \leq \lambda_D \leq \left(\frac{3}{2}F_n\right)^{1/n}.$$

Proof. The matrix formula gives $F_{2n} \geq 2F_n^2/n$. Thus $\log(F_n/n)/n$ is nondecreasing along dyadic orders and is bounded above by $\log 2$, since the signed-walk bound is $F_n \leq 2^{n-1}$. The difference between this quantity and $\log F_n/n$ tends to zero, proving existence. Complementation makes (r_i) palindromic. At even n, the sums over odd and even indices are each $F_n/2$, so $r_i \leq F_n/2$, $\sum r_i^2 \leq F_n^2/2$, and $\sum r_i r_{i+1} \leq F_n^2/4$. Therefore $F_{2n} \leq (3/2)F_n^2$. Iterating both inequalities and passing to the limit yields the bounds. □

The candidate growth constant need not equal the global growth constant unless further extremal information is proved.

3 Rigidity when the distance multiset is fixed

Theorem 3.1. Let $n = 2^k$. If a realizable word contains exactly 2^j copies of distance 2^j for every $0 \leq j < k$, then it is D_n.

Proof. Prove the stronger statement for Hamiltonian paths on the cyclic group $\mathbb{Z}/n\mathbb{Z}$, measuring an edge by its circular distance. The edges of distance $n/2$ form the antipodal perfect matching. A Hamiltonian path uses distinct matching edges, and there are exactly $n/2$ such edges in the prescribed multiset, so it uses them all. They cannot occupy neighboring edge positions. The only way to put $n/2$ nonneighboring positions among $n-1$ positions is to occupy positions $1, 3, \ldots, n-1$.

Contract each antipodal pair and reduce labels modulo $n/2$. The remaining path visits every contracted vertex once. Each other distance is at most $n/4$, so its circular distance is unchanged in the smaller cyclic group. Its multiset has exactly the required counts for $n/2$. Induction, starting with $n = 2$, forces the contracted word to be $D_{n/2}$; interleaving its entries with $n/2$ forces D_n.

For ordinary absolute differences in $[n]$, every prescribed distance is at most $n/2$ and hence equals the original circular distance. The stronger statement therefore applies. □

This proves uniqueness within a histogram class, not extremality: one must still show that a maximizer has that histogram. Reduction modulo $n/2$ must use the circular metric; ordinary absolute distances are not generally preserved by that reduction.

4 A computer-assisted theorem for one odd entry at order 32

For any 16-label word A, let $e_A(i)$ count realizations ending at i and put $\mathcal{E}(A) = \sum_i e_A(i)^2$. The archived exact computation establishes

$$\mathcal{E}(A) \le 13830, \qquad A \ne D_{16} \implies \mathcal{E}(A) \le 10590. \tag{1}$$

Its sole word at energy at least 10591 is D_{16}, with endpoint vector

$$u = (50, 36, 24, 35, 15, 12, 18, 25, 25, 18, 12, 15, 35, 24, 36, 50).$$

Certificate method and provenance. For unused set S and current label x, compute

$$U(\varnothing, x) = 1, \qquad U(S, x) = \max_d \sum_{\substack{y \in S \\ |y-x|=d}} U(S \setminus \{y\}, y).$$

Induction bounds the completions for every fixed suffix. At a word prefix retain all partial paths, including their multiplicities and initial labels. For initial label i, sum U over its paths to obtain b_i. Every extension's starting-count vector is componentwise bounded by b, hence its energy is at most $\sum b_i^2$. Prune only when this is below the fixed threshold 10591, and branch over all possible next distances otherwise. At a leaf the bound is exact. Starting-count and ending-count searches are equivalent under word reversal; D_{16} is palindromic.

The supplied archived files `threshold.cpp`, `halfstart_check.cpp`, and their completion logs record two implementations agreeing on the sole qualifying word and 318777320 visited prefix nodes. Admissible residual values obey $U \le 2^{15}$; the stored 16-bit values and 64-bit energy sums are sufficient. The energy-maximum log and the direct recount of u give 13830. These full-size runs were completed in the earlier project, not rerun for this manuscript; the source, logs and independent comparison are retained in `research/archive`. Fresh verification here recounts the dyadic endpoint vectors and checks the small permutation cases. Thus the following theorem explicitly depends on a finite computer certificate.

Theorem 4.1 (Restricted order-32 result)**.** Among all 32-label records having exactly one odd entry, D_{32} is the unique maximum-fiber word, with 48410 realizations.

Proof. A record with no realization has count zero. In any realization with a single odd jump, each side lies in one parity class and must exhaust its 16 labels; hence the odd entry is necessarily in position 16. The word is therefore $(2A, t, 2\,\mathrm{rev}\,B)$ for arbitrary 16-label words A, B and odd $t = 2r + 1$. Let $\mathrm{Adj}(s)$ have entries $\mathbf{1}_{|i-j|=s}$, with $\mathrm{Adj}(16) = 0$. Summing the two parity orders gives the exact count $e_A^{\mathsf{T}} K_r e_B$, where

$$K_0 = 2I + \mathrm{Adj}(1), \qquad K_r = \mathrm{Adj}(r) + \mathrm{Adj}(r+1) \quad (r \ge 1).$$

Every row and column sum is at most four, so $\|K_r\|_2 \le 4$. If either half differs from D_{16}, (1) bounds the integer count by

$$\left\lfloor \sqrt{16 \cdot 13830 \cdot 10590} \right\rfloor = 48408 < 48410.$$

If both halves equal D_{16}, direct integer evaluation of $u^{\mathsf{T}} K_r u$ for $t = 1, 3, \ldots, 31$ gives

$$48410, 37790, 33220, 28500, 24580, 25100, 25298, 23302,$$
$$20062, 17138, 17112, 19648, 17848, 14592, 12200, 5000.$$

Only $t = 1$ attains 48410, giving exactly D_{32}. □

5 A sharper bridge bound and a reusable certificate criterion

The row-sum bound four can be sharpened uniformly over every odd bridge. The improvement is useful because it increases the separation between a non-dyadic half and the candidate, without any larger search.

Proposition 5.1. For the $m \times m$ bridge matrices above, $m \geq 2$,

$$\|K_r\|_2 \leq \rho_m := 2 + 2\cos\frac{\pi}{m+1} \quad (r \geq 0).$$

Equality in this bound holds for $r = 0$; for $r \geq 1$ the bound is strict.

Proof. The adjacency matrix of a path with ℓ vertices has eigenvalues $2\cos(j\pi/(\ell+1))$, $1 \leq j \leq \ell$: substitute the vector with coordinates $\sin(ij\pi/(\ell+1))$ into the adjacency recurrence. Thus $K_0 = 2I + \mathrm{Adj}(1)$ has norm ρ_m.

For a positive distance s, $\mathrm{Adj}(s)$ is the disjoint union of paths on residue classes modulo s. The longest has $\lceil m/s \rceil$ vertices, giving

$$\|\,\mathrm{Adj}(s)\|_2 = 2\cos\frac{\pi}{\lceil m/s \rceil + 1}.$$

When $s \geq m$ this expression is zero, as required. For $r \geq 1$, the triangle inequality and the monotonicity in path length give

$$\|K_r\|_2 \leq 2\cos\frac{\pi}{m+1} + 2\cos\frac{\pi}{\lceil m/2 \rceil + 1} < \rho_m.$$

□

Proposition 5.2 (Improved separation at order 32)**.** Any one-odd-entry record at order 32 with at least one half different from D_{16} has at most 47996 realizations.

Proof. The archived energy gap (1) and Proposition 5.1 give the bound $\rho_{16}\sqrt{13830 \cdot 10590}$. For an entirely rational check, $\rho_{16} < 1983/500$. Indeed, $3.14159 < \pi < 3.14160$ and $\cos x \leq 1 - x^2/2 + x^4/24$ imply

$$\rho_{16} \leq 4 - \left(\frac{3.14159}{17}\right)^2 + \frac{1}{12}\left(\frac{3.14160}{17}\right)^4 < \frac{1983}{500}.$$

Squaring this rational upper bound and taking the integer floor yields

$$\left\lfloor \sqrt{\frac{1983^2 \cdot 13830 \cdot 10590}{500^2}} \right\rfloor = 47996.$$

All comparisons are checked using fractions and integer square roots in the new verification script. □

This leaves a separation of 414 from 48410, rather than relying on the row-sum estimate's separation of two. Conversely, a one-odd-entry competitor attaining 48410 would force each half-energy to be at least

$$\left\lceil \frac{48410^2 \cdot 500^2}{1983^2 \cdot 13830} \right\rceil = 10774.$$

The archived classification at the lower threshold 10591 is therefore more than sufficient for this step. The computation is still only on 16 labels; the spectral improvement changes the analytical comparison, not its coverage class.

More generally, suppose every m-label word has energy at most E, every word other than W has energy at most L, and a chosen odd bridge gives $T = e_W^{\mathsf{T}} K_{r_0} e_W$. If

$$T > \rho_m \sqrt{EL}$$

and all other bridges between two copies of W give strictly less than T, then $(2W, 2r_0 + 1, 2\operatorname{rev} W)$ is uniquely optimal among *all one-odd-entry* records on $2m$ labels. The proof is exactly the two-case argument above. This is a reusable sufficient criterion; obtaining its energy hypotheses at larger m is a separate unresolved problem.

6 Parity-run balance beyond a single crossing

Let a realizable record on $2m$ labels have q odd entries, and let $\ell_1, \ldots, \ell_{q+1}$ be the lengths of its constant-parity vertex runs. Necessarily

$$\sum_{j \text{ odd}} \ell_j = \sum_{j \text{ even}} \ell_j = m, \qquad \ell_j \geq 1.$$

Counting positive compositions gives the exact number of possible run-length vectors:

$$\begin{cases} \binom{m-1}{t-1}^2, & q = 2t-1, \\ \binom{m-1}{t}\binom{m-1}{t-1}, & q = 2t. \end{cases}$$

Each such vector is a possible parity pattern: fill its odd-run positions with the m labels of one parity and the other positions with the rest. This does not guarantee a realization for an independently prescribed distance word.

At order 32, $q = 1, 2, 3, 4, 5$ allow respectively $1, 15, 225, 1575, 11025$ run-length vectors. For one odd jump the unique vector is $(16, 16)$, which explains the endpoint-energy reduction. For two, the middle run is forced to have length 16; the next section exploits this fact. With three odd jumps both parity classes are split, so a full-parity Hamiltonian-path matrix by itself no longer describes either side. One needs counts of disjoint partial paths and their correlations. This is a precise obstruction to extending the one-crossing proof merely by repeating its scalar norm bound.

7 Two odd entries: an exact reduction without a bound

A path on 32 labels with two odd jumps has three parity runs of sizes $(a, 16, 16-a)$, where $1 \leq a \leq 15$. Write its word as

$$d = (2A, 2r+1, 2B, 2s+1, 2C),$$

where the lengths of A, B, C are $a-1, 15, 15-a$. Let $E_B(i, j)$ count full 16-label realizations of B with specified first and last labels. Let $P_{A,C}(u, v)$ count ordered pairs of disjoint paths

covering [16], whose words are A, C, with the first path ending at u and the second starting at v; the other two endpoints are free. Finally set

$$C_r(i,j) = \mathbf{1}_{\{i-j=-r \text{ or } r+1\}}.$$

With $\langle X, Y\rangle = \sum_{u,v} X_{uv}Y_{uv}$, a direct path decomposition gives

$$A_{32}(d) = \left\langle P_{A,C},\ C_r E_B C_s^{\mathsf{T}} + C_r^{\mathsf{T}} E_B C_s \right\rangle. \tag{2}$$

Indeed, when the outer runs are odd labels, the two bridge compatibilities give the first product; when the outer runs are even, they give the second. The disjoint outer paths and the middle full path uniquely reconstruct a realization. The supplied `verify_parity_reductions.py` checks the analogous identity against every record at orders 4 and 8, including all 1079 two-odd-entry words at order 8.

Although the matrices have only 256 entries, their feasible families are large and correlated. Bounding the three factors independently loses too much information. Nor is arbitrary gluing automatically a positive semidefinite form: already at order four, $A_4(3,2,1)^2 = 4$ whereas $A_4(3,2,3)A_4(1,2,1) = 0$. Thus a generic self-gluing Cauchy–Schwarz argument is false.

8 Exact boundary of current knowledge

Complete fresh enumeration proves the conjectured maximum and uniqueness at $n = 2, 4, 8$. The archived complete search gives the same at $n = 16$, and its maximizing witness was freshly recounted. Theorems 3.1 and 4.1 cover specific classes at larger orders. They do not show that an unrestricted maximizing 32-label word has one odd jump or the prescribed dyadic histogram. Words with at least two odd entries remain the central missing class, and (2) has no sufficient uniform bound yet. At order 64, even the candidate count alone gives no global comparison. The remaining task is a structural domination theorem or a rigorous upper bound covering *all* competitors, not a larger calculation of the same candidate.

Provenance and use of AI

This is a research draft, not a peer-reviewed publication. The problem arose in an AI-assisted exploratory project. AI tools were used substantially for literature searches, mathematical derivations, verification programs, and drafting. The proofs and bounded checks are supplied for independent scrutiny; neither an exhaustive novelty claim nor a general proof beyond the stated results is asserted.

References

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